

Multiple Choice

1. **Answer:** D

Options: A) Partitioning based clustering; B) K-means clustering; C) Grid based clustering; D) All of the above

Note: Correct option: D - All of the above

2. **Answer:** A

Options: A) anomaly detection; B) one-class detection; C) train-test mismatch robustness; D) background detection

Note: Correct option: A - anomaly detection

3. **Answer:** C

Options: A) The use of a non-Gaussian noise model in probabilistic regression.; B) The use of probabilistic modelling for regression.; C) The use of prior distributions on the parameters in a probabilistic model.; D) The use of class priors in Gaussian Discriminant Analysis.

Note: Correct option: C - The use of prior distributions on the parameters in a probabilistic model.

4. **Answer:** A

Options: A) $P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)$; B) $P(X, Y, Z) = P(X) * P(Y|X) * P(Z|Y)$; C) $P(X, Y, Z) = P(Z) * P(X|Z) * P(Y|Z)$; D) $P(X, Y, Z) = P(X) * P(Y) * P(Z)$

Note: Correct option: A - $P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)$

5. **Answer:** A

Options: A) $p(y|x, w)$; B) $p(y, x)$; C) $p(w|x, w)$; D) None of the above

Note: Correct option: A - $p(y|x, w)$

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'True, True'.

9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def extract_min_max(test_tup, K): | return (res)`
Note: Students should provide a detailed explanation referencing algorithm steps.
12. **Answer:** `def heap_ascending(nums): | return s_result`
Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key